

SMART PLACEMENT PREPARATION PORTAL

Prepared in the partial fulfillment of the Summer Internship Program on AWS AT



Under the guidance of

Mrs. Sumana Bethala, APSSDC

Mr. Lovababu Juttuka, APSSDC

Submitted by

TIRUMANADHAM MURALI KRISHNA, 24225A4507, BVC ODALAREVU, (Batch -1)

KAPILESWARAPU PRASAD, 23221A4521, BVC ODALAREVU, (Batch -1)

UDDISAPU RAMCHARAN, 24225A4508, BVC ODALAREVU, (Batch -1)

SUNKARA AMBIKA NAGA SRIVALLI, 24225A4505, BVC ODALAREVU, (Batch -1)

KURASALA SRI NAGA LAKSHMI PARVATHI, 24225A4502, BVC ODALAREVU, (Batch -1)

DEVAGUPTAPU PAVANI, 25221DBA07, BVC ODALAREVU, (Batch -1)

CHAVVAKULA PURNIMA SAI LAKSHMI, 25221DBA05, BVC ODALAREVU, (Batch -1)

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to all those who have contributed to the successful completion of my summer internship project at **Andhra Pradesh Skill Development Corporation (APSSDC)**. This opportunity has been an enriching and transformative experience for me, and I am truly thankful for the support, guidance, and encouragement I have received along the way.

First and foremost, I extend my sincere regards to Mrs Sumana Bethala and Mr Lovababu juttuka, my supervisors and mentors, for providing me with valuable insights, constant guidance, and unwavering support throughout the duration of the internship. Their expertise and encouragement have been instrumental in shaping the direction of this project.

I would like to thank the entire team at **Andhra Pradesh Skill Development Corporation (APSSDC)** for fostering a collaborative and innovative environment. The camaraderie, knowledge sharing, and feedback I received from my colleagues significantly contributed to the development and success of this project.

In conclusion, I am honoured to have been a part of this internship program, and I look forward to leveraging the skills and knowledge gained to contribute positively to future endeavours.

Thank you.

Sincerely,

Tirumanadham Murali Krishna, 24225A4507, 24225a4507@bvcgroup.in
Kapileswarapu Prasad, 23221A4521, ksdp015@gmail.com
Uddisapu Ramcharan, 24225A4508, uddisapuramcharan@gmail.com
Sunkara Ambika Naga Srivalli, 24225A4505, 24225a4505@bvcgroup.in
Kurasala Sri Naga Lakshmi Parvathi, 24225A4502, 24225a4502@bvcgroup.in
Devaguptapu Pavani, 25221DBA07, pavanid044@gmail.com
Chavvakula Purnima Sai Lakshmi, 25221DBA05, purnimachavvakula16@gmail.com

ABSTRACT

The **Smart Placement Preparation Portal** is a **cloud-based web application** developed to provide students with a **centralized platform for placement preparation**. The project integrates **aptitude practice, coding resources, core computer science subjects, resume building, mock interviews, company-specific preparation, and placement updates** into a single application, enabling students to prepare efficiently through a structured learning environment.

The application is developed using **React.js** and **Vite** for the frontend, **Spring Boot** for the backend, and **PostgreSQL** for database management. It is deployed on **Amazon Web Services (AWS)** using an **Amazon EC2 (Elastic Compute Cloud)** instance running **Ubuntu Linux**. To ensure secure and reliable access, services such as **Elastic IP, Virtual Private Cloud (VPC), Security Groups, and Nginx** are utilized. The project is managed using **Git** and **GitHub**, while deployment and remote server management are performed through **SSH, VS Code Remote SSH, and WinSCP**.

This project demonstrates the practical implementation of **full-stack web development, cloud computing, and DevOps practices** through real-world AWS deployment. It provided hands-on experience in **application development, cloud deployment, Linux server administration, version control, and collaborative software development**. Overall, the **Smart Placement Preparation Portal** provides a **scalable, secure, and user-friendly solution** that simplifies placement preparation while strengthening our technical knowledge and practical skills in **modern software development and cloud technologies**.

TABLE OF CONTENTS

S.No	Content	Pg.No
1.	INTRODUCTION	5
2.	METHODOLOGY	6
3.	TECHNOLOGY STACK	9
4.	SYSTEM DESIGN / ARCHITECTURE	12
5.	IMPLEMENTATION	19
6.	RESULTS	23
7.	CONCLUSION	29

1. INTRODUCTION

The rapid advancement of technology has transformed the way students prepare for campus placements. Success in placement drives requires knowledge of aptitude, programming, data structures, core computer science subjects, resume building, mock interviews, and company-specific preparation. However, students often depend on multiple websites and applications to access these resources, making the learning process time-consuming, unorganized, and difficult to manage. Switching between different platforms can reduce productivity and affect the overall learning experience.

To overcome these challenges, we developed the **Smart Placement Preparation Portal**, a **cloud-based web application** that provides a **centralized platform for placement preparation**. The portal integrates essential learning resources into a single application, enabling students to access **aptitude practice, coding resources, core computer science subjects, AI & Emerging Technologies, resume building, mock tests, interview preparation, community discussions, and placement updates** through a simple and user-friendly interface. This integrated approach simplifies the preparation process and helps students learn in a structured and efficient manner.

The application is developed using **React.js** and **Vite** for the frontend, **Spring Boot** for the backend, and **PostgreSQL** for database management. To gain practical exposure to **cloud computing** and **DevOps practices**, the application is deployed on **Amazon Web Services (AWS)** using an **Amazon EC2 (Elastic Compute Cloud)** instance running **Ubuntu Linux**. Services such as **Elastic IP, Virtual Private Cloud (VPC), Security Groups, and Nginx** are configured to provide secure, reliable, and continuous access to the application. The project is

managed using **Git** and **GitHub**, while deployment and remote server management are performed through **SSH**, **VS Code Remote SSH**, and **WinSCP**.

The project provided valuable hands-on experience in **full-stack web development**, **cloud computing**, **Linux server administration**, and **DevOps practices**. It enhanced our understanding of **application deployment**, **cloud infrastructure management**, **version control**, **collaborative software development**, and real-world project implementation. Overall, the **Smart Placement Preparation Portal** serves as a **scalable, secure, and user-friendly solution** that supports students in achieving their placement goals while strengthening our technical knowledge and practical skills.

2. METHODOLOGY

The development of the Smart Placement Preparation Portal followed a structured and systematic methodology to ensure efficient development, secure deployment, and reliable application performance. The methodology was divided into several phases, including requirements gathering, system design, implementation, database management, testing, deployment, and iterative refinement. Each phase was carefully planned to develop a scalable, secure, and user-friendly web application using modern full-stack development technologies and Amazon Web Services (AWS). The following subsections describe each phase of the project in detail.

2.1 Requirements Gathering

The project began by analyzing the challenges faced by students during placement preparation. Many students rely on multiple platforms for aptitude practice, coding resources, interview preparation, resume building, and placement updates, making the learning process unorganized and time-consuming. Based on these requirements, the objective was to develop the **Smart Placement Preparation Portal**, a centralized platform that provides all essential placement preparation resources through a single web application. The functional and technical requirements were finalized after identifying user needs and project objectives.

2.2 System Design

Based on the gathered requirements, the overall architecture of the Smart Placement Preparation Portal was designed using a modern full-stack approach. The application consists of three major components: the frontend, backend, and database. The frontend was developed using React.js and Vite, while Spring Boot was used to develop the backend services. PostgreSQL was selected as the database, and Amazon Web Services (AWS) was chosen as the deployment platform to ensure scalability, security, and reliable application hosting.

2.3 Implementation

The implementation phase focused on developing the core functionalities of the application. Various modules such as Aptitude & Reasoning, Coding & Data Structures, Core Computer Science, AI & Emerging Technologies, Resume Builder, Mock Tests, Interview Preparation, Community Discussions, and Placement Updates were integrated into the portal. REST APIs were implemented to establish communication between the frontend and backend, while Git and GitHub were used for version control and collaborative development throughout the project lifecycle.

2.4 Database Management

PostgreSQL was used as the relational database management system to store and manage user information, learning resources, mock tests, placement details, and other application data. Proper database design and validation techniques were implemented to ensure data consistency, integrity, and efficient retrieval of information. The database was successfully integrated with the backend to provide secure and reliable data management.

2.5 Testing

Multiple levels of testing were performed to ensure the proper functioning of the Smart Placement Preparation Portal. Functional testing verified the working of user authentication, navigation, learning modules, and other core features. Database testing ensured successful data storage and retrieval from PostgreSQL, while deployment testing confirmed that the application was accessible through the AWS EC2 public IP. Security testing was also performed to ensure that only authorized network traffic was allowed through AWS Security Groups.

2.6 Deployment

After successful development and testing, the Smart Placement Preparation Portal was deployed on Amazon Web Services (AWS) using an Amazon EC2 instance running Ubuntu Linux. Nginx was configured as the web server, while an Elastic IP was associated with the EC2 instance to provide a static public IP address. Security Groups and Virtual Private Cloud (VPC) were configured to ensure secure network communication. Remote deployment and server management were carried out using SSH, VS Code Remote SSH, and WinSCP, enabling the application to be securely hosted and accessed through the public IP address.

2.7 Iterative Refinement

Throughout the development process, continuous improvements were made based on testing results, team discussions, and mentor feedback. Minor issues related to the user interface, application performance, module integration, and deployment were resolved to improve usability, stability, and overall system performance. This iterative approach helped deliver a reliable and user-friendly application.

Methodology Summary

The methodology adopted for the development of the Smart Placement Preparation Portal ensured the successful design, development, testing, and deployment of a secure, scalable, and user-friendly web application. By integrating React.js, Vite, Spring Boot, PostgreSQL, AWS EC2, Nginx, Ubuntu Linux, Git, and GitHub, the project demonstrates the practical implementation of full-stack web development, cloud computing, and DevOps practices in building a modern placement preparation platform

3. TECHNOLOGY STACK

The **Smart Placement Preparation Portal** was developed using a modern full-stack technology stack that provides a responsive user interface, secure backend services, reliable database management, and cloud-based deployment. The project utilizes **React.js** and **Vite** for the frontend, **Spring Boot** for backend development, **PostgreSQL** for database management, and **Amazon Web Services (AWS)** for hosting and deployment. The following technologies and services were used throughout the implementation of the project.

3.1 Front-End Technology

- **React.js:** Used to develop a responsive and interactive user interface through reusable components, providing an enhanced user experience.
- **Vite:** Used as the frontend build tool to enable faster development, quick module replacement, and optimized application performance.
- **HTML5, CSS3, and JavaScript:** Used to design and develop the user interface with responsive layouts and interactive web pages.

3.2 Back-End Technology

- **Spring Boot:** Used to develop RESTful APIs and implement the business logic of the application. It provides secure, scalable, and efficient backend services for communication between the frontend and the database.

3.3 Database

- **PostgreSQL:** Used as the relational database management system to securely store user information, learning resources, placement updates, mock test details, resume information, and other application data. It provides high performance, reliability, and data integrity.

3.4 AWS Cloud Services

- **Amazon EC2 (Elastic Compute Cloud):** Used to host the application on a virtual server, providing a reliable and scalable cloud computing environment.
- **Virtual Private Cloud (VPC):** Provides a secure networking environment for hosting cloud resources and managing communication between AWS services.
- **Security Groups:** Configured as virtual firewalls to control inbound and outbound network traffic, ensuring secure access to the application.

- **Elastic IP:** Used to assign a static public IP address to the EC2 instance, ensuring consistent access to the deployed application.

3.5 Web Server

- **Nginx:** Configured as a web server and reverse proxy to efficiently serve the React frontend and forward backend requests to the Spring Boot application, improving application performance and reliability.

3.6 Operating System

- **Ubuntu Linux:** Used as the operating system for the Amazon EC2 instance. It provides a stable, secure, and developer-friendly environment for application deployment and server management.

3.7 Deployment & Remote Access

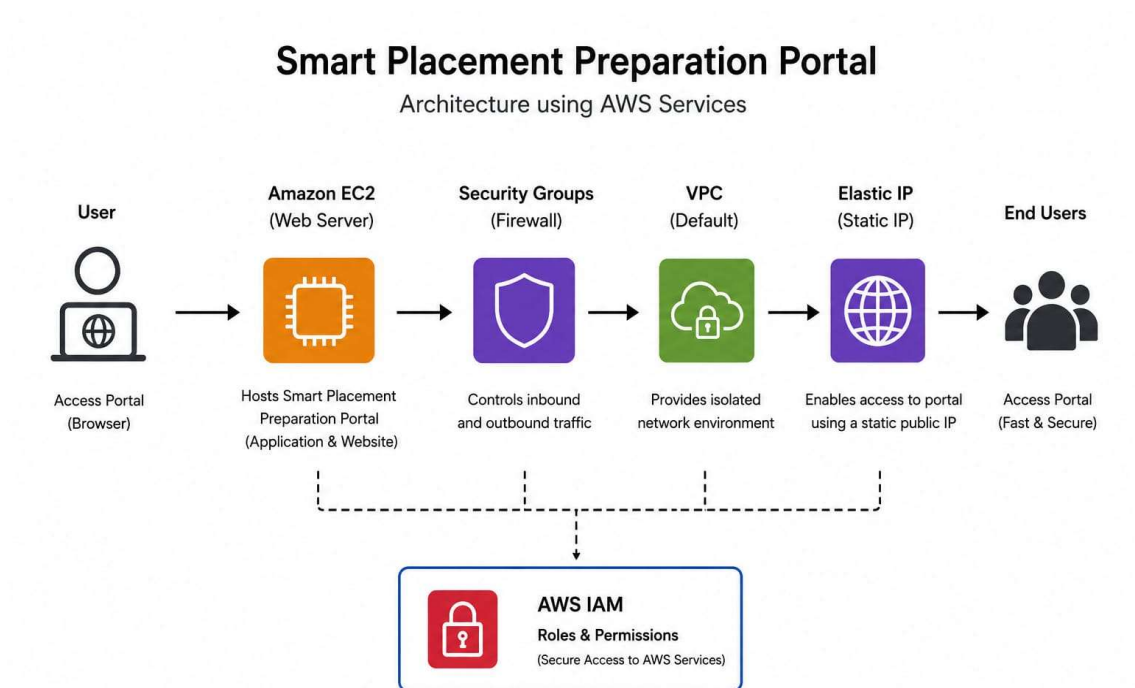
- **SSH (Secure Shell):** Used to securely connect to the Ubuntu EC2 instance for server administration and deployment.
- **VS Code Remote SSH:** Used to remotely access and manage the cloud server directly from Visual Studio Code, simplifying development and maintenance.
- **WinSCP:** Used to securely transfer project files between the local system and the EC2 server during deployment and configuration.

3.8 Version Control

- **Git:** Used to manage source code changes, track project history, and support collaborative software development.
- **GitHub:** Used as the remote repository to store the project source code, maintain version control, and facilitate collaboration among team members.

4. SYSTEM DESIGN / ARCHITECTURE

The **Smart Placement Preparation Portal** is designed using **Amazon Web Services (AWS)** to provide a secure, scalable, and reliable cloud environment. The application is hosted on an **Amazon EC2** instance, while **Security Groups** and a **Virtual Private Cloud (VPC)** ensure secure network communication. An **Elastic IP** provides a static public IP address for consistent access, and **AWS Identity and Access Management (IAM)** manages secure access to AWS resources. This architecture enables students to access the portal efficiently through a stable and secure cloud infrastructure.



4.1 Web Server Layer – Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) is used to host the **Smart Placement Preparation Portal**. It provides a secure, scalable, and reliable virtual server environment for deploying the Spring Boot application and handling user requests efficiently. The EC2 instance

acts as the primary hosting platform where the backend application processes requests from users. It offers high availability, flexible resource allocation, and easy scalability to support future application growth. By hosting the application on EC2, the portal ensures stable performance, secure access, and reliable service for both students and administrators.

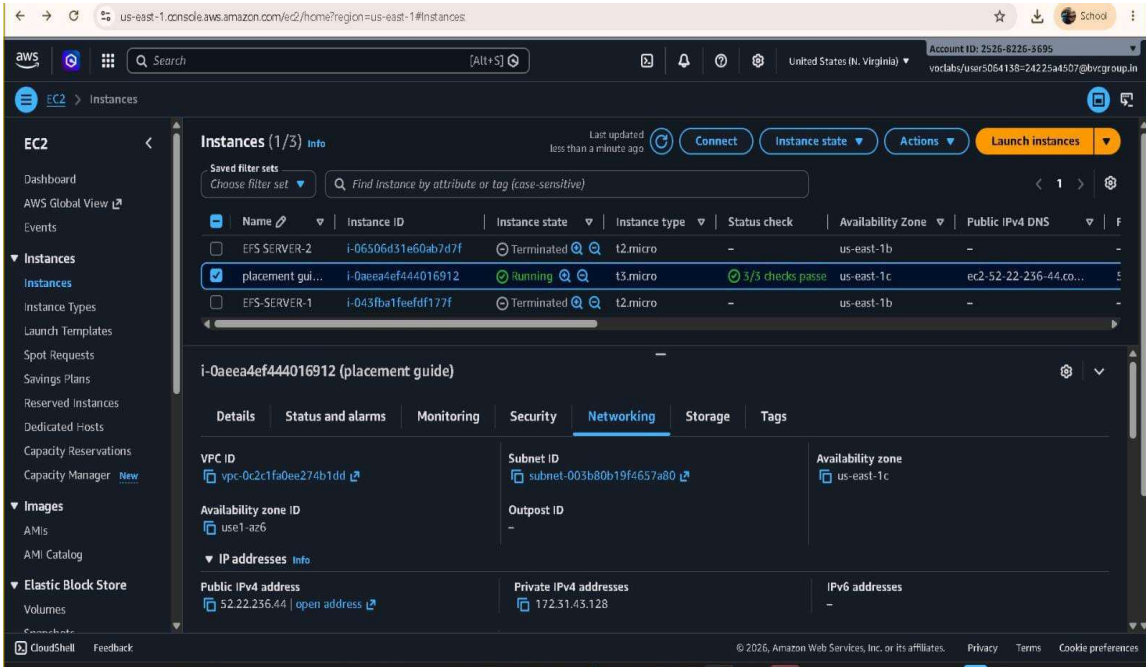
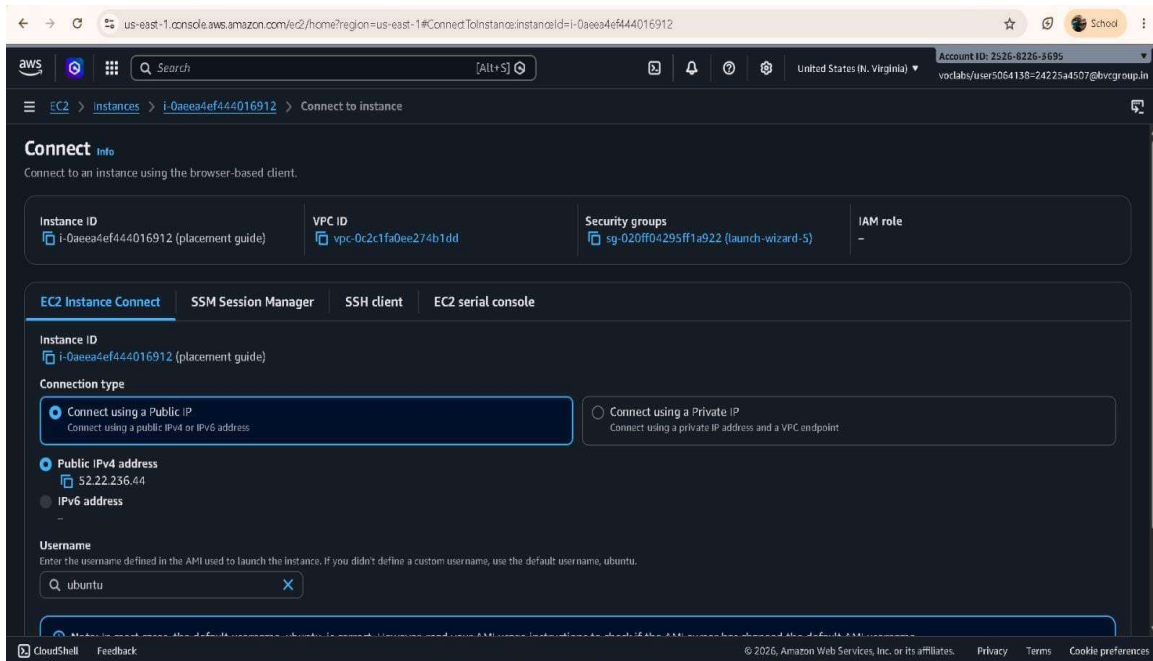


Figure 4.1: Amazon EC2 Instance Hosting the Smart Placement Preparation Portal

4.2 Remote Server Access – EC2 Instance Connect

Amazon EC2 Instance Connect is used to securely access the EC2 instance through a web browser without requiring third-party SSH tools. It enables administrators to connect to the server, execute commands, deploy applications, and perform maintenance tasks efficiently. This browser-based connection simplifies server management while maintaining secure access

to the application environment. It also supports quick troubleshooting and monitoring of the deployed application.



4.2: EC2 Instance Connect for secure remote server access.

4.3 Application Deployment on Amazon EC2

The **Smart Placement Preparation Portal** is deployed on the Amazon EC2 instance using the Spring Boot framework. After deployment, the application initializes all required components, including the embedded Tomcat server and database connection, ensuring that the backend services run successfully. The deployment process enables the portal to handle user requests efficiently while maintaining high availability and reliable performance. This cloud-based deployment supports seamless access to the application from different locations.

```
aws
Search [Alt+S]
United States (N. Virginia)
Account ID: 2526-8226-3695
voclabs/user5064138=24225a4507@bvggroup.in

ubuntu@ip-172-31-43-128: ~/Smart-Placement-Portal/simple-backend$ java -jar target/portal-1.0.0.jar

:: Spring Boot ::
(v3.2.0)

2026-07-05T14:52:39.006Z INFO 2421 --- [main] c.p.portal.PlacementPortalApplication : Starting PlacementPortalApplication v1.0.0 using Java 21.0.11 with P
ID 2421 (/home/ubuntu/Smart-Placement-Portal/simple-backend/target/portal-1.0.0.jar started by ubuntu in /home/ubuntu/Smart-Placement-Portal/simple-backend)
2026-07-05T14:52:39.012Z INFO 2421 --- [main] c.p.portal.PlacementPortalApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T14:52:40.439Z INFO 2421 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-07-05T14:52:40.567Z INFO 2421 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 88 ms. Found 3 JPA repos
itory interfaces.
2026-07-05T14:52:41.404Z INFO 2421 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-05T14:52:41.419Z INFO 2421 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T14:52:41.419Z INFO 2421 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.16]
2026-07-05T14:52:41.471Z INFO 2421 --- [main] o.s.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-05T14:52:41.473Z INFO 2421 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2278 ms
2026-07-05T14:52:41.725Z INFO 2421 --- [main] o.hibernate.jpa.internal.util.LogHelper : HHH0000204: Processing PersistenceUnitInfo [name: default]
2026-07-05T14:52:41.830Z INFO 2421 --- [main] org.hibernate.Version : HHH0000412: Hibernate ORM core version 6.3.1.Final
2026-07-05T14:52:41.893Z INFO 2421 --- [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-07-05T14:52:42.290Z INFO 2421 --- [main] o.s.c.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-07-05T14:52:42.391Z INFO 2421 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-05T14:52:42.561Z INFO 2421 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@4e0
53cb9
2026-07-05T14:52:42.563Z INFO 2421 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-07-05T14:52:44.684Z INFO 2421 --- [main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH0000489: No JTA platform available (set 'hibernate.transaction.jta
.platform' to enable JTA platform integration)
2026-07-05T14:52:44.747Z INFO 2421 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'

i-0aeaa4ef444016912 (placement guide)
PublicIPs: 52.22.236.44 PrivateIPs: 172.31.43.128

CloudShell Feedback
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
```

4.3: Successful deployment of the Smart Placement Preparation Portal on Amazon EC2.

4.4 Network Layer – Amazon Virtual Private Cloud (VPC)

Amazon Virtual Private Cloud (VPC) provides a secure and isolated networking environment for the Smart Placement Preparation Portal. The EC2 instance is deployed inside the VPC, allowing secure communication between AWS resources while protecting the application from unauthorized network access. The VPC offers better network management, improved security, and controlled connectivity for cloud resources. It also ensures reliable communication within the AWS infrastructure.

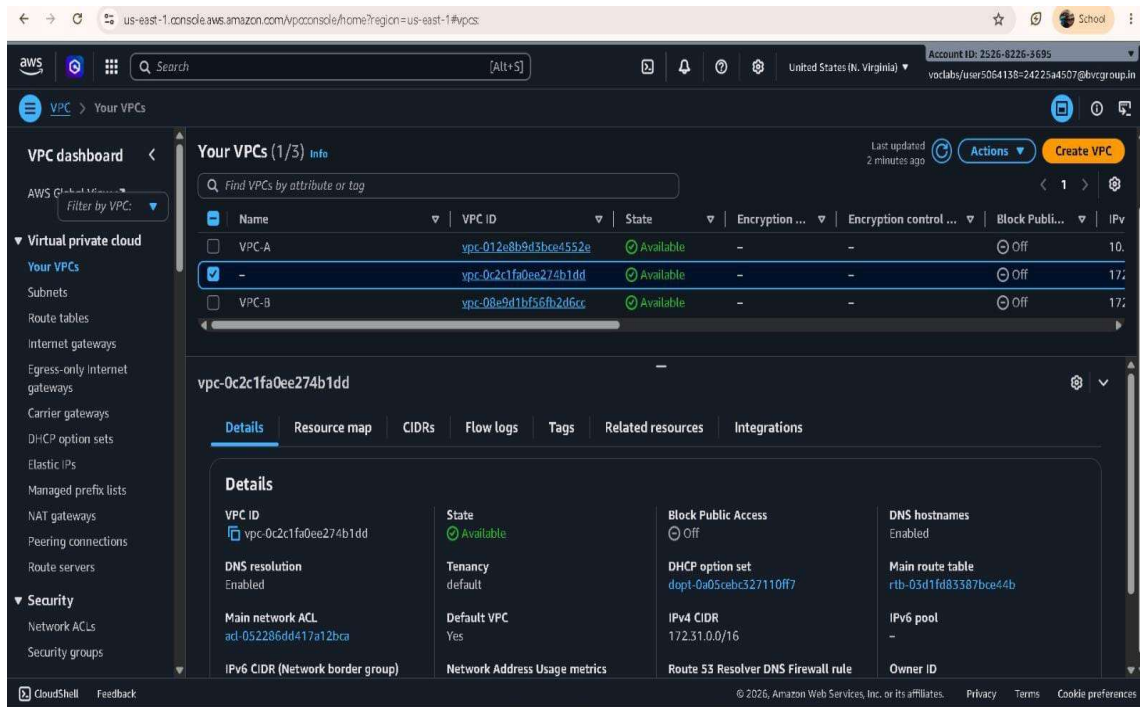
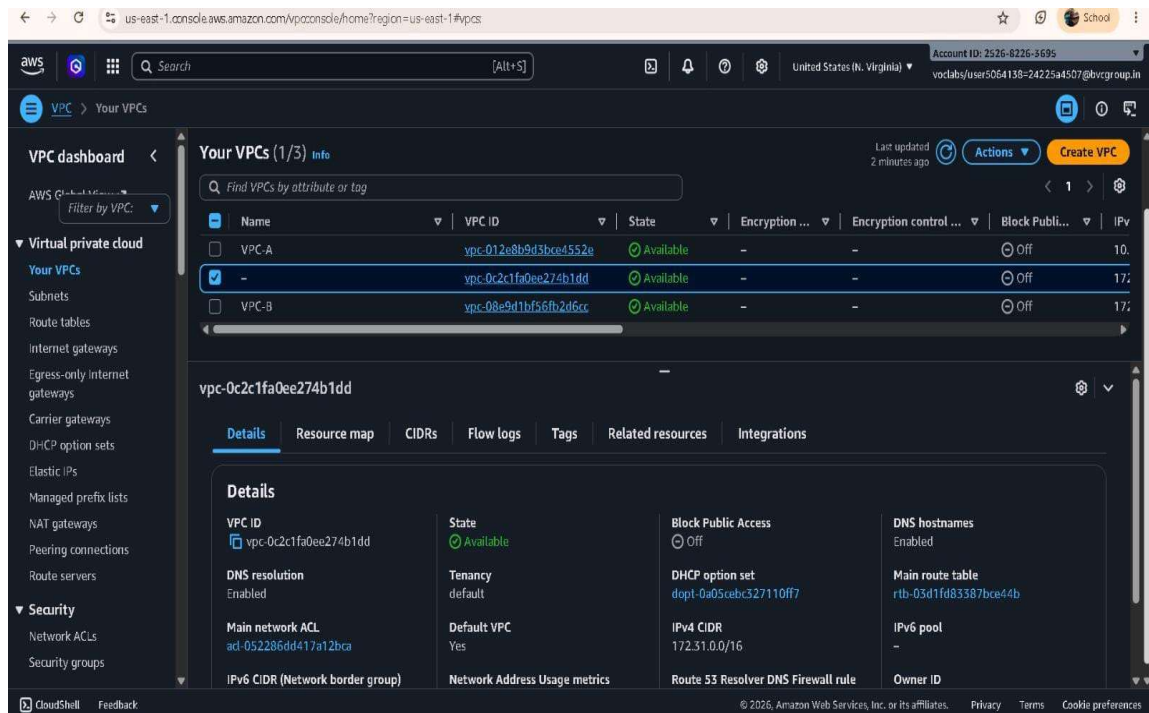


Figure 4.4: Amazon Virtual Private Cloud (VPC) configuration.

4.5 Security Layer – AWS Security Groups

Amazon Virtual Private Cloud (Amazon VPC) provides a secure and isolated networking environment for the Smart Placement Preparation Portal. The EC2 instance is deployed within the VPC, allowing secure communication between AWS resources while preventing unauthorized network access. It enables controlled network configuration, IP address management, and secure connectivity for the hosted application. By using Amazon VPC, the application achieves improved security, reliability, and efficient network management.



4.4: Amazon Virtual Private Cloud (VPC) configuration for the Smart Placement Preparation Portal.

4.5 Security Layer – AWS Security Groups

AWS Security Groups act as a virtual firewall for the Amazon EC2 instance by controlling both inbound and outbound network traffic. They allow only authorized users and services to access the Smart Placement Preparation Portal while blocking unauthorized connections. Security Groups enhance the overall security of the application by enforcing access rules and protecting the server from potential network threats. This ensures secure communication between the EC2 instance and other AWS resources.

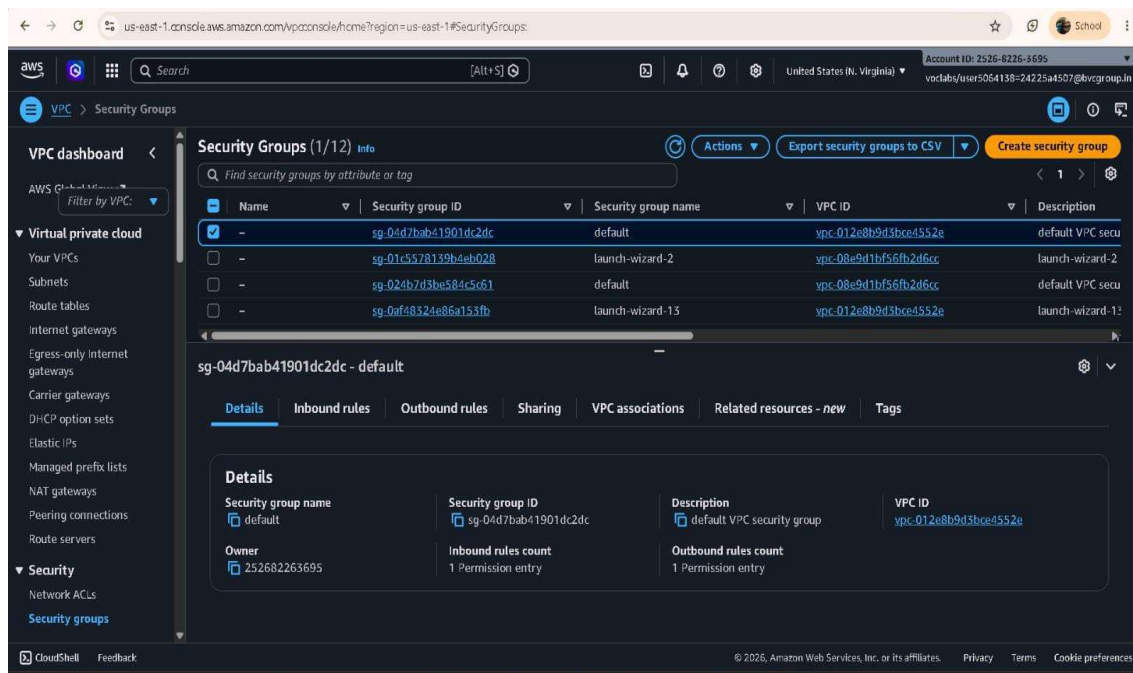


Figure 4.5: AWS Security Groups configuration for the Smart Placement Preparation Portal.

4.6 Public Access Layer – Elastic IP Address

An Elastic IP address is associated with the Amazon EC2 instance to provide a static public IP address for the Smart Placement Preparation Portal. This ensures that users can access the application through a consistent public endpoint even after instance restarts. Elastic IP improves application availability, simplifies remote access, and provides a reliable connection for cloud-hosted services. It also supports stable deployment and efficient management of the application.

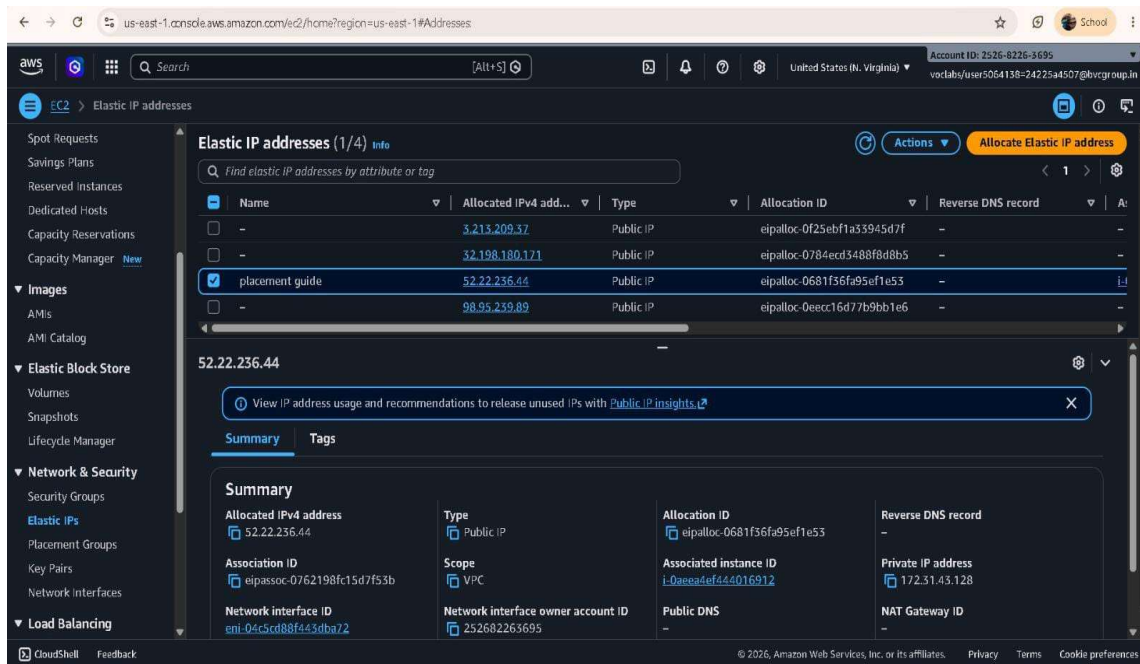


Figure 4.6: Elastic IP associated with the Amazon EC2 instance.

5. IMPLEMENTATION

The implementation of the Smart Placement Preparation Portal transforms the proposed system design into a fully functional web application deployed on the AWS cloud platform. The application is developed using Java Spring Boot and hosted on an Amazon EC2 instance. AWS networking and security services are configured to ensure secure deployment, reliable connectivity, and efficient application performance. This chapter explains the implementation process of each major component involved in deploying and running the Smart Placement Preparation Portal.

5.1 Deployment of Smart Placement Preparation Portal on Amazon EC2

The implementation process begins by launching an Amazon EC2 instance, which serves as the hosting environment for the Smart Placement Preparation Portal. The Spring Boot application

is packaged as an executable JAR file and transferred to the EC2 instance. After deployment, the application is executed using the Java Runtime Environment, enabling the server to host the portal and process user requests efficiently.

Implementation Steps:

1. Launch an Amazon EC2 instance.
2. Connect to the server using EC2 Instance Connect.
3. Upload the Spring Boot application to the EC2 instance.
4. Build and package the Spring Boot application using Maven.
5. Execute the application using the `java -jar` command.

5.2 Running the Spring Boot Application

After deployment, the Spring Boot application is executed on the EC2 instance. The embedded Tomcat server starts automatically and initializes all application components. The application establishes a connection with the PostgreSQL database and loads all required backend services before accepting user requests.

Application Execution Process:

- Starts the embedded Apache Tomcat server.
- Initializes Spring Boot components.
- Establishes the database connection.
- Loads REST APIs and application services.
- Accepts incoming user requests.

5.3 Database Integration

The Smart Placement Preparation Portal uses a PostgreSQL relational database to store and manage application data. Student details, administrator information, study materials, mock tests, placement updates, and user authentication records are securely maintained within the database. Spring Boot communicates with the database using Spring Data JPA to perform efficient CRUD (Create, Read, Update, Delete) operations.

Database Functions:

- Stores student registration details.
- Maintains administrator information.
- Manages study materials and mock tests.
- Stores placement updates.
- Performs CRUD operations using Spring Data JPA.

5.4 Network Configuration

Amazon Virtual Private Cloud (VPC) is configured to provide a secure networking environment for the Smart Placement Preparation Portal. The EC2 instance is deployed within the VPC, allowing secure communication between cloud resources while isolating the application from unauthorized network access. Elastic IP is associated with the EC2 instance to provide a stable public IP address for accessing the application.

Network Configuration:

- Configures Amazon VPC for network isolation.
- Provides secure communication between AWS resources within the Virtual Private Cloud (VPC).
- Associates an Elastic IP with the EC2 instance.

- Enables secure internet connectivity.

5.5 Security Configuration

AWS Security Groups are configured as virtual firewalls to protect the Smart Placement Preparation Portal. They regulate inbound and outbound traffic by allowing only authorized ports required for server administration and application access. This security mechanism prevents unauthorized access while ensuring secure communication with the hosted application.

Security Configuration:

- Allows SSH access through Port 22.
- Allows application access through Port 8081.
- Restricts unauthorized network traffic.
- Protects the hosted application from external threats.

5.6 Application Verification

After completing the deployment and configuration process, the Smart Placement Preparation Portal is verified to ensure successful operation. The application starts without errors, establishes the required database connection, and becomes accessible through the configured public IP address. The successful implementation demonstrates the integration of Java Spring Boot with AWS cloud services to provide a secure, scalable, and reliable placement preparation platform.

Verification Results:

- Application deployed successfully on Amazon EC2.
- Spring Boot server started successfully.
- Database connectivity established.
- Users can access the application through the public IP address.
- The portal is ready for student and administrator operations.

6. RESULTS

The Smart Placement Preparation Portal was successfully developed and deployed on Amazon Web Services (AWS). The application was tested to verify its functionality, accessibility, cloud deployment, and overall performance. The results demonstrate that the system provides a secure, scalable, and user-friendly platform that supports students in preparing for campus placements through a centralized learning environment.

6.1 Successful AWS Deployment

The Smart Placement Preparation Portal was successfully deployed on an Amazon EC2 instance running Ubuntu Linux. The application was accessible through the configured Elastic IP, confirming successful cloud deployment. The Spring Boot backend and supporting services executed successfully, enabling reliable application hosting on the AWS platform.

Functional User Interface.

6.2 Functional User Interface

The portal provides a responsive and interactive user interface that enables students to navigate different sections easily. Modules such as **Aptitude & Reasoning, Coding & Data Structures, Core Computer Science, AI & Emerging Technologies, Resume Builder, Mock Tests, Interview Preparation, Community Discussions, and Placement Updates** were successfully displayed and tested. Navigation between modules functioned smoothly without errors.

6.3 Successful Database Integration

The **PostgreSQL** database was successfully integrated with the **Spring Boot** backend. User information, learning resources, placement details, and other application data were stored and retrieved efficiently. Database operations were performed successfully, ensuring reliable data management throughout the application.

6.4 Cloud Infrastructure Performance

The AWS infrastructure, including **Amazon EC2, Virtual Private Cloud (VPC), Security Groups, and Elastic IP**, performed successfully throughout deployment and testing. The configured cloud environment provided secure communication, stable application performance, and reliable access for users.

6.5 Secure Access and Connectivity

Secure access to the application was maintained using **AWS Security Groups, Virtual Private Cloud (VPC), and SSH**. Only authorized network traffic was allowed to access the application,

ensuring secure communication between cloud resources and protecting the deployed environment.

6.6 Overall System Performance

The Smart Placement Preparation Portal performed successfully during testing and deployment. The application responded efficiently to user requests, provided smooth navigation across all learning modules, and demonstrated reliable cloud performance. The implemented AWS architecture ensures scalability, security, and availability, making the portal suitable for future enhancements and real-world deployment.

6.7 Screenshots of the Output

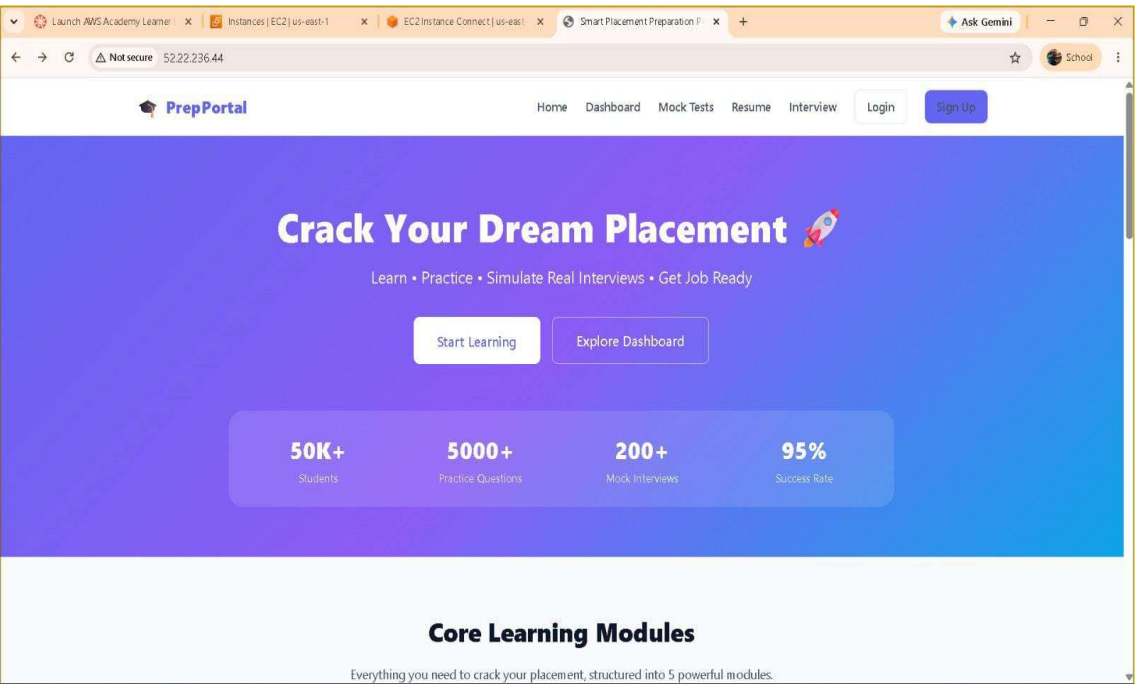


Figure 6.1: Home Page of Smart Placement Preparation Portal

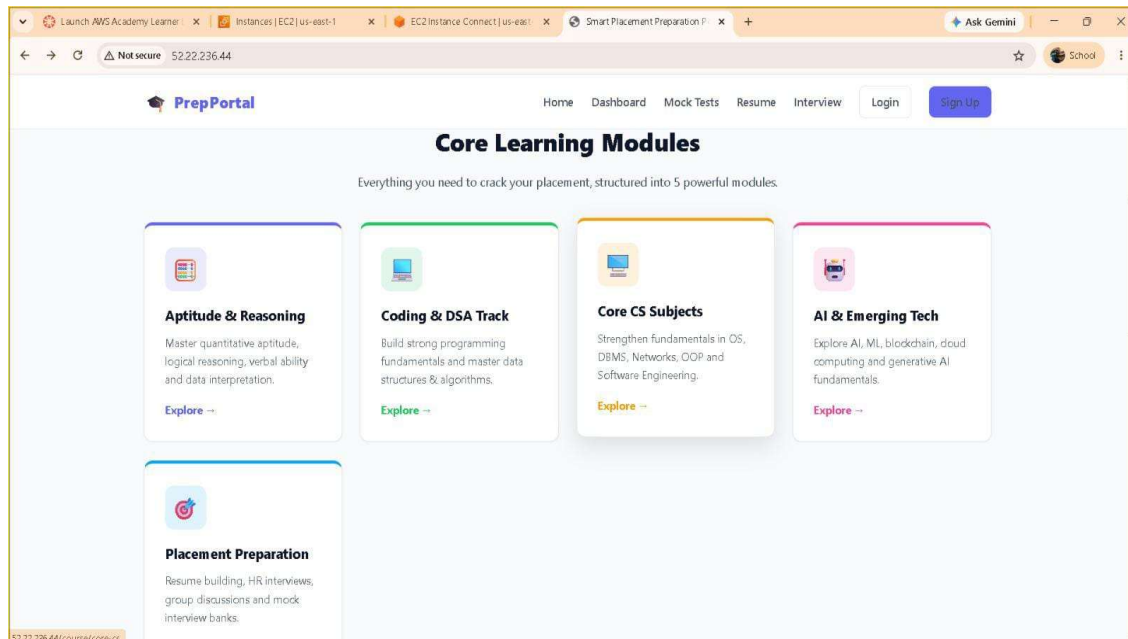


Figure 6.2: Core Learning Modules of the Smart Placement Preparation Portal.

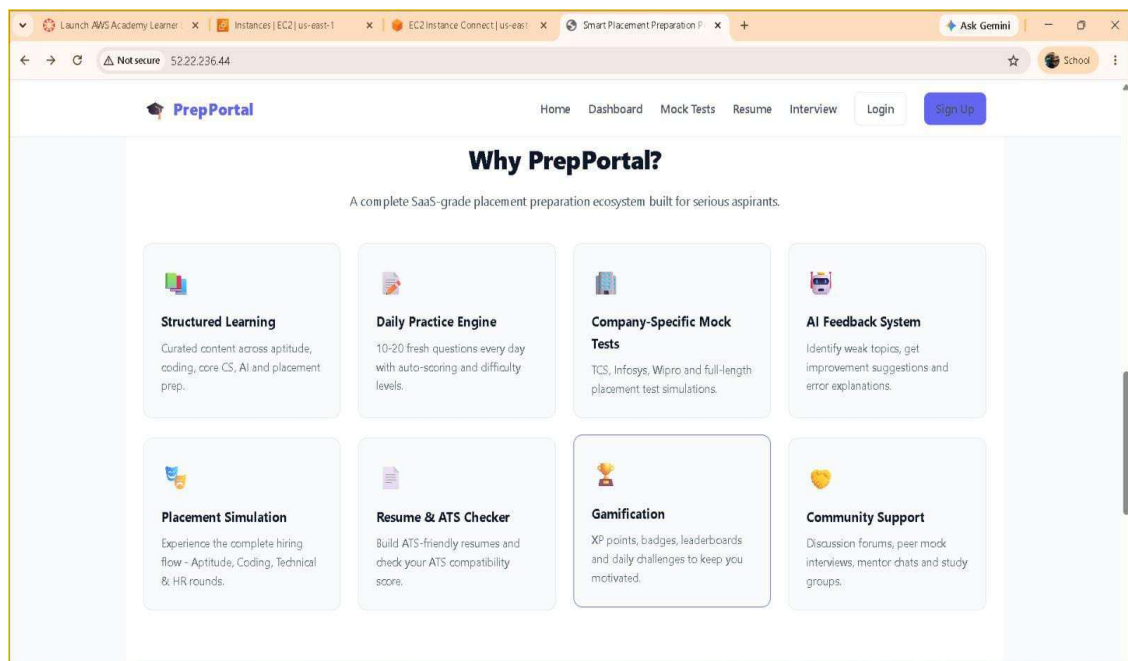


Figure 6.3: Features of the Smart Placement Preparation Portal.

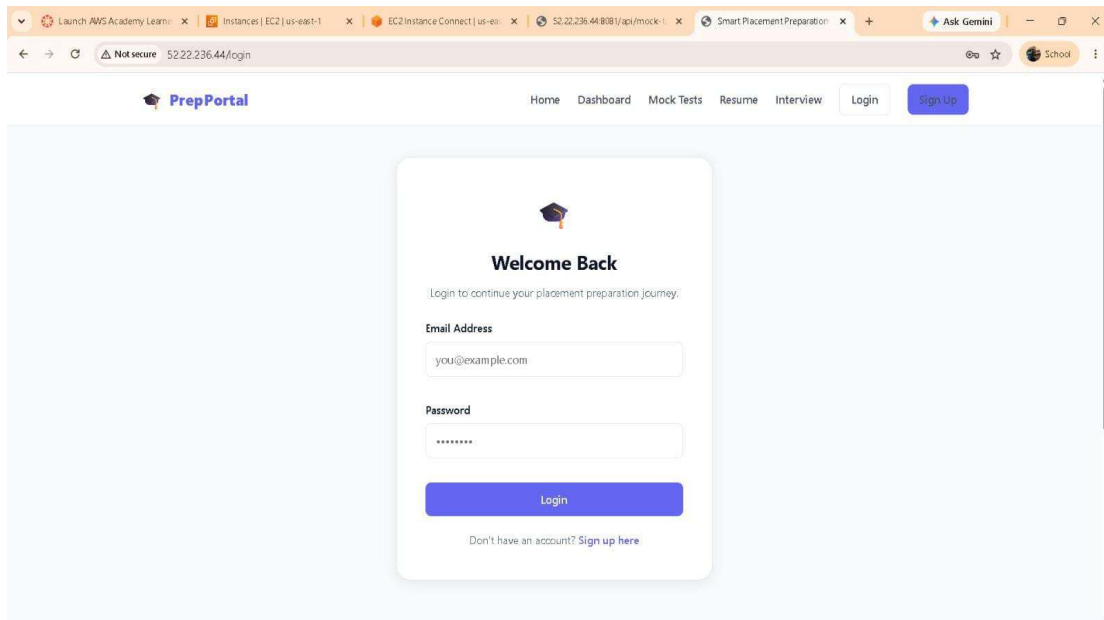


Figure 6.4: User Login Interface of the Smart Placement Preparation Portal

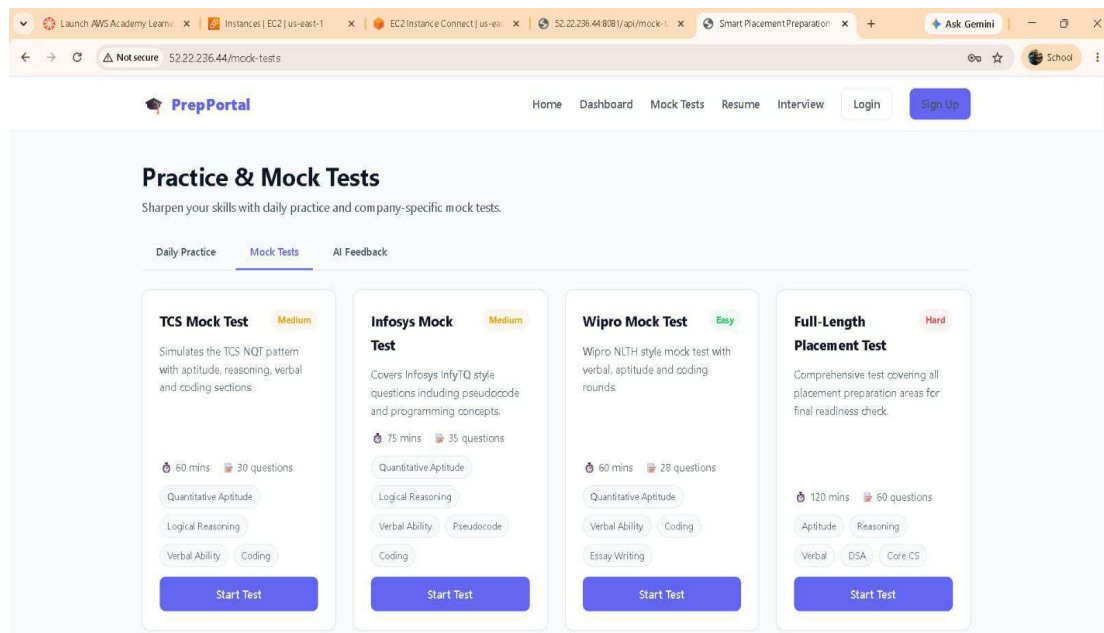


Figure 6.4: Mock Tests Module of the Smart Placement Preparation Portal

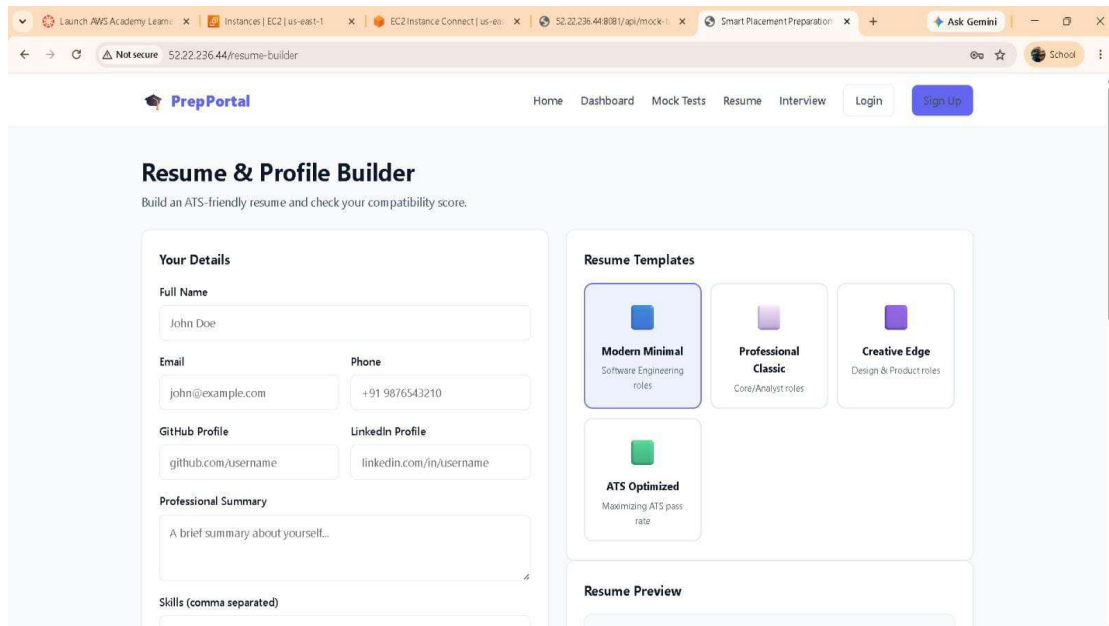


Figure 6.5: Resume Builder Module of the Smart Placement Preparation Port

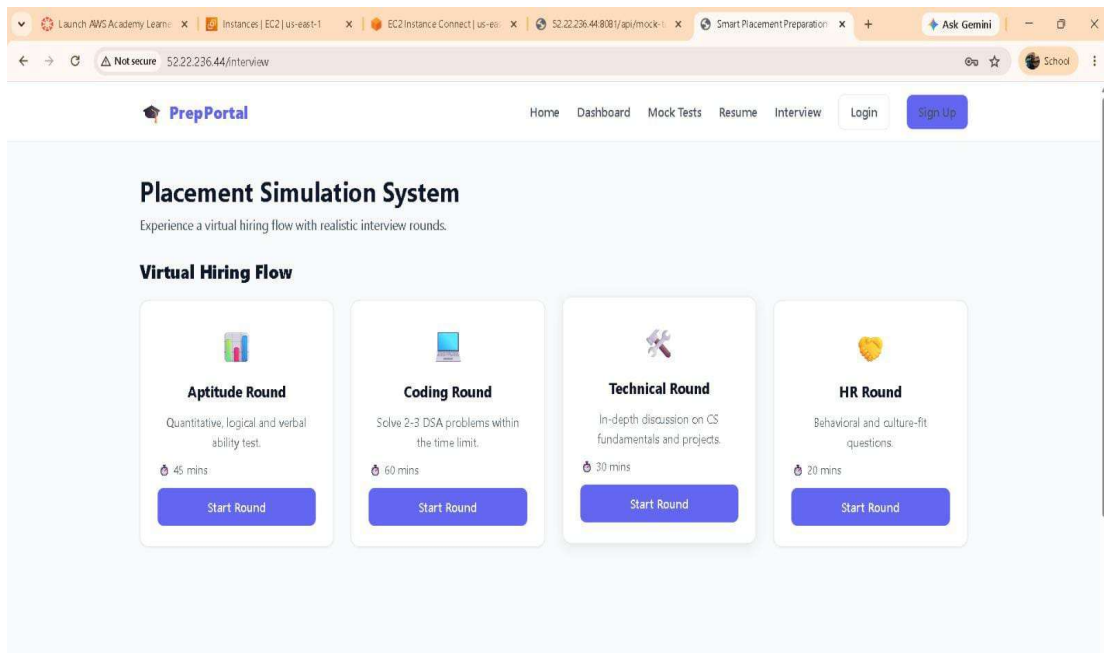


Figure 6.6: Placement Simulation System of the Smart Placement Preparation Portal

7.CONCLUSION

The **Smart Placement Preparation Portal** was successfully designed, developed, and deployed as a cloud-based web application to provide students with a centralized platform for placement preparation. The project successfully achieved its objective of integrating aptitude practice, coding resources, core computer science subjects, AI and emerging technologies, resume building, mock tests, interview preparation, community discussions, and placement updates into a single, user-friendly application. This integrated approach simplifies the placement preparation process and enables students to access all essential learning resources through one platform.

The application was developed using **React.js** and **Vite** for the frontend, **Spring Boot** for the backend, and **PostgreSQL** for database management. It was successfully deployed on **Amazon Web Services (AWS)** using an **Amazon EC2** instance with **Virtual Private Cloud (VPC)**, **Security Groups**, **Elastic IP**, and **Nginx**, ensuring secure, reliable, and scalable application hosting. **Git**, **GitHub**, **SSH**, **VS Code Remote SSH**, and **WinSCP** were used for version control, deployment, and server management.

Throughout the project, we gained valuable practical experience in **full-stack web development**, **cloud computing**, **Linux server administration**, **application deployment**, and **DevOps practices**. The implementation of AWS services enhanced our understanding of real-world cloud infrastructure, secure networking, and modern software deployment techniques. The project also strengthened our technical, problem-solving, and teamwork skills through collaborative development and deployment activities.

Overall, the Smart Placement Preparation Portal provides a secure, scalable, and user-friendly solution for placement preparation. It successfully demonstrates the practical implementation of modern web technologies and AWS cloud services while helping students prepare effectively for campus recruitment. The successful completion of this project enhanced our practical knowledge of software development, cloud computing, and DevOps practices, while strengthening our technical expertise, teamwork, and problem-solving skills for future professional challenges.